

Electrical and Electronic Measurements

Previous Year Questions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electrical and Electronic Measurements (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

1

Error Analysis

Topics

1. Static Characteristics of Measuring Device
2. Types of Error and Uncertainties of error

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Unit Snapshot*

Stream: EE	**Questions:** 13	**MCQ:** 7	**MSQ:** 0	**NAT:** 6
1M: 9	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 NAT 2025 1M

The maximum percentage error in the equivalent resistance of two parallel-connected resistors of 100 Ω and 900 Ω with each having a maximum 5% error is _____ % (round off to nearest integer value).

Q1.22.1 MCQ 2022 1M

A balanced Wheatstone bridge $ABCD$ has the following arm resistances: $R_{AB} = 1\,\text{k}\Omega \pm 2.1\%$; $R_{BC} = 100\,\Omega \pm 0.5\%$; R_{CD} is an unknown resistance; $R_{DA} = 300\,\Omega \pm 0.4\%$. The value of R_{CD} and its accuracy is

A. $30\,\Omega \pm 3\,\Omega$ **B.** $30\,\Omega \pm 0.9\,\Omega$
C. $3000\,\Omega \pm 90\,\Omega$ **D.** $3000\,\Omega \pm 3\,\Omega$

Q1.20.1 NAT 2020 2M

A non-ideal Si-based pn junction diode is tested by sweeping the bias applied across its terminals from −5 V to +5 V. The effective thermal voltage, V_T, for the diode is measured to be (29 ± 2) mV. The resolution of the voltage source in the measurement range is 1 mV. The percentage uncertainty (rounded off to 2 decimal places) in the measured current at a bias voltage of 0.02 V is _____.

Q1.17.1 NAT 2017 1M

The following measurements are obtained on a single phase load: $V = 220\,\text{V} \pm 1\%$, $I = 5.0\,\text{A} \pm 1\%$ and $W = 555\,\text{W} \pm 2\%$. If the power factor is calculated using these measurements, the worst case error in the calculated power factor in percent is (Give answer up to one decimal place.)

Q1.17.2 MCQ 2017 1M

Two resistors with nominal resistance values R_1 and R_2 have additive uncertainties ΔR_1 and ΔR_2, respectively. When these resistances are connected in parallel, the standard deviation of the error in the equivalent resistance R is

A. $\pm\sqrt{\left\{\frac{\partial R}{\partial R_1}\Delta R_1\right\}^2 + \left\{\frac{\partial R}{\partial R_2}\Delta R_2\right\}^2}$

B. $\pm\sqrt{\left\{\frac{\partial R}{\partial R_2}\Delta R_1\right\}^2 + \left\{\frac{\partial R}{\partial R_1}\Delta R_2\right\}^2}$

C. $\pm\sqrt{\left\{\frac{\partial R}{\partial R_1}\right\}^2 \Delta R_2 + \left\{\frac{\partial R}{\partial R_2}\right\}^2 \Delta R_1}$

D. $\pm\sqrt{\left\{\frac{\partial R}{\partial R_1}\right\}^2 \Delta R_1 + \left\{\frac{\partial R}{\partial R_2}\right\}^2 \Delta R_2}$

Q1.15.1 NAT 2015 1M

A capacitive voltage divider is used to measure the bus voltage V_{bus} in a high-voltage 50 Hz AC system as shown in the figure. The measurement capacitors C_1 and C_2 have tolerances of ±10% on their nominal capacitance values. If the bus voltage V_{bus} is 100 kV rms, the maximum rms output voltage V_{out} (in kV), considering the capacitor tolerances, is

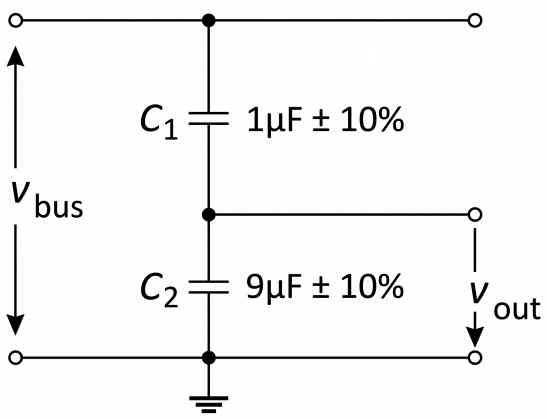


Fig. Q1.15.1

Q1.06.1 MCQ 2006 2M

A variable w is related to three other variables x, y, z as $w = xy/z$. The variables are measured with meters of accuracy ±0.5% reading, ±1% of full scale value and ±1.5% reading. The actual readings of the three meters are 80, 20 and 50 with 100 being the full scale value for all three. The maximum uncertainty in the measurement of w will be

A. ±0.5% rdg **B.** ±5.5% rdg
C. ±6.7% rdg **D.** ±7.0% rdg

Q1.99.1 MCQ 1999 2M

Two $100\,\mu A$ full scale PMMC meters are employed to construct a $10\,V$ and a $100\,V$ full scale voltmeter. These meters will have figures of merit (sensitivities) as

A. $10\,k\Omega/V$ and $10\,k\Omega/V$ **B.** $100\,k\Omega/V$ and $10\,k\Omega/V$
C. $10\,k\Omega/V$ and $100\,k\Omega/V$ **D.** $10\,k\Omega/V$ and $1\,k\Omega/V$

Q1.97.1 NAT 1997 2M

In the circuit shown in fig. For measuring resistance 'R' if the ammeter indicates 1 A and the voltmeter indicate 100 V, then the value of R is _____ Ohms and the error in measurement using the ratio V/I is _____ %.

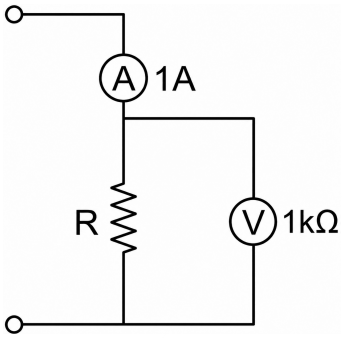


Fig. Q1.97.1

Q1.95.1 NAT 1995 1M

Fringing in capacitive type transducer can be minimized by providing a _____

Q1.95.2 MCQ 1995 1M

Four ammeters M1, M2, M3 and M4 with the following specifications are available.

Instrument	Type	Full scale value (A)	Accuracy % of FS
M1	$3\frac{1}{2}$ digit dual slope	20	±0.10
M2	PMMC	10	±0.20
M3	Electro-dynamic	5	±0.50
M4	Moving-iron	1	±1.00

Fig. Q1.95.2

A current of 1 A is to be measured. To obtain minimum error in the reading, one should select meter

A. M1 **B.** M2
C. M3 **D.** M4

Q1.94.1 MCQ 1994 1M

A precise measurement guarantees accuracy of the measured quantity.

A. TRUE

B. FALSE

Q1.92.1 MCQ 1992 1M

A resistance is measured by the voltmeter- ammeter method employing d.c. excitation and a voltmeter of very high resistance connected directly across the unknown resistance. If the voltmeter and ammeter readings are subject to maximum possible errors of ±2.4% and ±1.0% respectively, then the magnitude of the maximum possible percentage error in the value of resistance deduced from the measurement is nearly

A. 1.4% **B.** 1.7% **C.** 2.4% **D.** 3.4%

Answer Key

Q1.25.1 5 to 5	Q1.22.1 B	Q1.20.1 5 to 6
Q1.17.1 4.0 to 4.1	Q1.17.2 A	Q1.15.1 11.75 to 12.25
Q1.06.1 D	Q1.99.1 A	Q1.97.1 111.11, 10
Q1.95.1 Guard ring	Q1.95.2 D	Q1.94.1 B
Q1.92.1 D		

UNIT

II

Analog Measuring Instruments

Topics

1. PMMC Type Instruments
2. Moving Iron & Dynamometer Type Instruments
3. Dynamometer Type Wattmeter
4. LPF Wattmeter
5. Two Wattmeter Measurement
6. Energy Meter
7. Hot wire Instruments & Electrostatic Voltmeters

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Unit Snapshot*

Stream: EE	**Questions:** 65	**MCQ:** 49	**MSQ:** 0	**NAT:** 16
1M: 37	**2M:** 25	**Descriptive:** 3	**Avg Weightage** ~3M	**Range:** 1M(2020)–4M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q2.24.1 NAT 2024 2M

In the circuit shown, $Z_1 = 50\angle -90^\circ\ \Omega$ and $Z_2 = 200\angle -30^\circ\ \Omega$. It is supplied by a three phase 400 V source with the phase sequence being R-Y-B. Assume the watt meters W_1 and W_2 to be ideal. The magnitude of the difference between the readings of W_1 and W_2 in watts is _____ (rounded off to 2 decimal places).

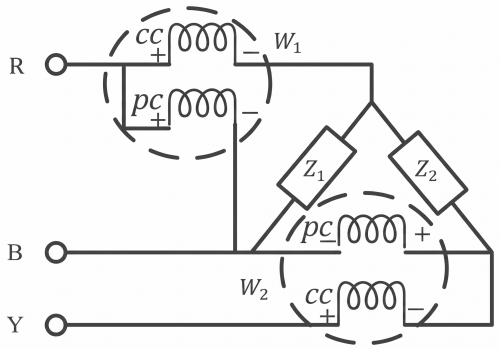


Fig. Q2.24.1

Q2.23.1 MCQ 2023 2M

A 3-phase, star-connected, balanced load is supplied from a 3-phase, 400 V (rms), balanced voltage source with phase sequence R-Y-B, as shown in the figure. If the wattmeter reading is −400 W and the line current is $I_R = 2$ A (rms), then the power factor of the load per phase is

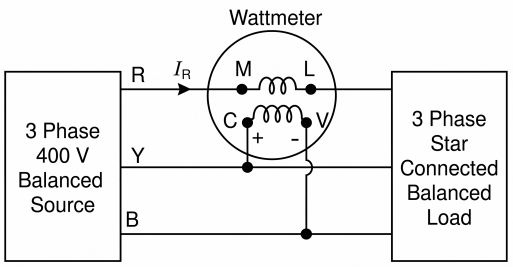


Fig. Q2.23.1

A. Unity
B. 0.5 leading
C. 0.866 leading
D. 0.707 lagging

Q2.20.1 NAT 2020 1M

Currents through ammeters A2 and A3 in the figure are $1\angle 10^\circ$ and $1\angle 70^\circ$, respectively. The reading of the ammeter A1 (rounded off to 3 decimal places) is _____ A.

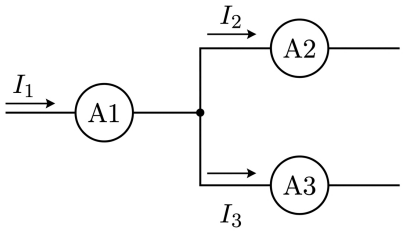


Fig. Q2.20.1

Q2.19.1 MCQ 2019 2M

A moving coil instrument having a resistance of 10 Ω, gives a full-scale deflection when the current is 10 mA. What should be the value of the series resistance, so that it can be used as a voltmeter for measuring potential difference up to 100 V?

A. 9 Ω **B.** 99 Ω **C.** 990 Ω **D.** 9990 Ω

Q2.18.1 MCQ 2018 1M

Two wattmeter method is used for measurement of power in a balanced three-phase load supplied from a balanced three-phase system. If one of the wattmeters reads half of the other (both positive), then the power factor of the load is

A. 0.532 **B.** 0.632 **C.** 0.707 **D.** 0.866

Q2.18.2 MCQ 2018 2M

A 0-1 Ampere moving iron ammeter has an internal resistance of 50 mΩ and inductance of 0.1 mH. A shunt coil is connected to extend its range to 0-10 Ampere for all operating frequencies. The time constant in milliseconds and resistance in mΩ of the shunt coil respectively are

A. 2, 5.55 **B.** 2, 1 **C.** 2.18, 0.55 **D.** 11.1, 2

Q2.17.1 MCQ 2017 2M

The load shown in the figure is supplied by a 400 V (line to line) 3-phase source (RYB sequence). The load is balanced and inductive, drawing 3464 VA. When the switch S is in position N, the three watt-meters W_1, W_2 and W_3 read 577.35 W each. If the switch is moved to position Y, the readings of the watt-meters in watts will be:

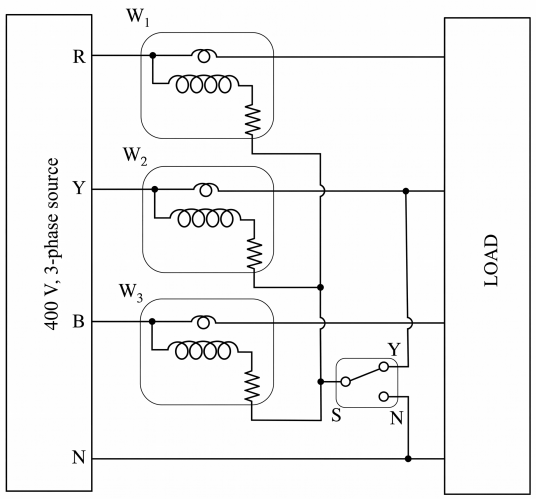


Fig. Q2.17.1

A. $W_1 = 1732$ and $W_2 = W_3 = 0$

B. $W_1 = 0$, $W_2 = 1732$ and $W_3 = 0$

C. $W_1 = 866$, $W_2 = 0$, $W_3 = 866$

D. $W_1 = W_2 = 0$ and $W_3 = 1732$

Q2.16.1 NAT 2016 2M

A dc voltage with ripple is given by $v(t) = [100+10\sin(\omega t)-5\sin(3\omega t)]$ volts. Measurements of this voltage $v(t)$, made by moving-coil and moving-iron voltmeters, show readings of V_1 and V_2 respectively. The value of $V_2 - V_1$, in volts, is _____.

Q2.16.2 NAT 2016 2M

An energy meter, having meter constant of 1200 revolutions/kWh, makes 20 revolutions in 30 seconds for a constant load. The load, in kW, is _____.

Q2.15.1 NAT 2015 1M

A (0 – 50 A) moving coil ammeter has a voltage drop of 0.1 V across its terminals at full scale deflection. The external shunt resistance (in milliohms) needed to extend its range to (0 – 500 A) is _____.

Q2.15.2 MCQ 2015 1M

Match the following.

Instrument Type	Used for
P. Permanent magnet moving coil	1. DC Only
Q. Moving iron connected through current transformer	2. AC Only
R. Rectifier	3. AC and DC
S. Electrodynamometer	

A. P-1, Q-2, R-1, S-3 **B.** P-1, Q-3, R-1, S-2
C. P-1, Q-2, R-3, S-3 **D.** P-3, Q-1, R-2, S-1

Q2.15.3 MCQ 2015 1M

A 3-phase balanced load which has a power factor of 0.707 is connected to a balanced supply. The power consumed by the load is 5 kW. The power is measured by the two-wattmeter method. The readings of the two wattmeters are

A. 3.94 kW and 1.06 kW **B.** 2.50 kW and 2.50 kW
C. 5.00 kW and 0.00 kW **D.** 2.96 kW and 2.04 kW

Q2.15.4 NAT 2015 2M

The coils of a wattmeter have resistances 0.01 Ω and 1000 Ω; their inductances may be neglected. The wattmeter is connected as shown in the figure, to measure the power consumed by a load, which draws 25 A at power factor 0.8. The voltage across the load terminals is 30 V. The percentage error on the wattmeter reading is _____.

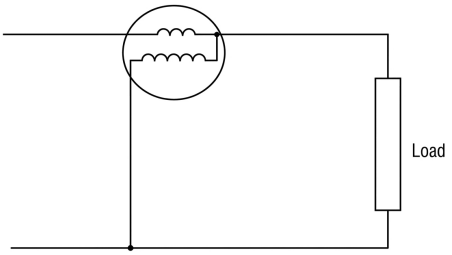


Fig. Q2.15.4

Q2.14.1 NAT 2014 1M

An LPF wattmeter of power factor 0.2 is having three voltage settings 300 V, 150 V and 75 V, and two current settings 5 A and 10 A. The full scale reading is 150. If the wattmeter is used with 150 V voltage setting and 10 A current setting, the multiplying factor of the wattmeter is _____.

Q2.14.2 NAT 2014 1M

While measuring power of a three-phase balanced load by the two-wattmeter method, the readings are 100 W and 250 W. The power factor of the load is _____.

Q2.14.3 MCQ 2014 1M

Power consumed by a balanced 3-phase, 3-wire load is measured by the two wattmeter method. The first wattmeter reads twice that of the second. Then the load impedance angle in radians is

A. $\pi/12$ **B.** $\pi/8$ **C.** $\pi/6$ **D.** $\pi/3$

Q2.14.4 MCQ 2014 2M

A periodic waveform observed across a load is represented by

$$V(t) = \begin{cases} 1 + \sin \omega t & 0 \le \omega t < 6\pi \\ -1 + \sin \omega t & 6\pi \le \omega t < 12\pi \end{cases}$$

The measured value, using moving iron voltmeter connected across the load, is

A. $\sqrt{\frac{3}{2}}$ **B.** $\sqrt{\frac{2}{3}}$ **C.** $\frac{3}{2}$ **D.** $\frac{2}{3}$

Q2.14.5 NAT 2014 2M

Two ammeters X and Y have resistances of 1.2 Ω and 1.5 Ω respectively and they give full-scale deflection with 150 mA and 250 mA respectively. The ranges have been extended by connecting shunts so as to give full scale deflection with 15 A. The ammeters along with shunts are connected in parallel and then placed in a circuit in which the total current flowing is 15 A. The current in amperes indicated in ammeter X is _____.

Q2.14.6 NAT 2014 1M

The saw-tooth voltage waveform shown in the figure is fed to a moving iron voltmeter. Its reading would be close to _____.

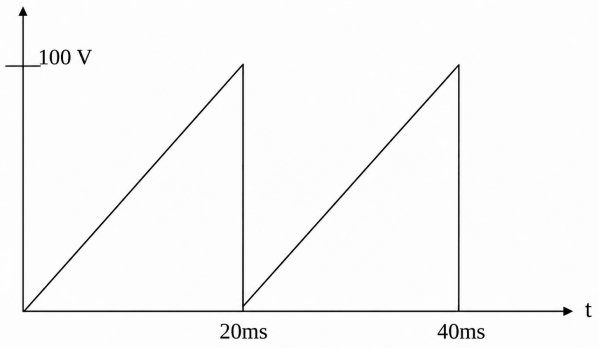


Fig. Q2.14.6

Q2.14.7 NAT 2014 2M

The dc current flowing in a circuit is measured by two ammeters, one PMMC and another electrodynamometer type, connected in series. The PMMC meter contains 100 turns in the coil, the flux density in the air gap is 0.2 Wb/m^2, and the area of the coil is 80 mm^2. The electrodynamometer ammeter has a change in mutual inductance with respect to deflection of 0.5 mH/deg. The spring constants of both the meters are equal. The value of current, at which the deflections of the two meters are same, is _____.

Q2.13.1 MCQ 2013 1M

The input impedance of the permanent magnet moving coil (PMMC) voltmeter is infinite. Assuming that the diode shown in the figure below is ideal, the reading of the voltmeter in Volts is

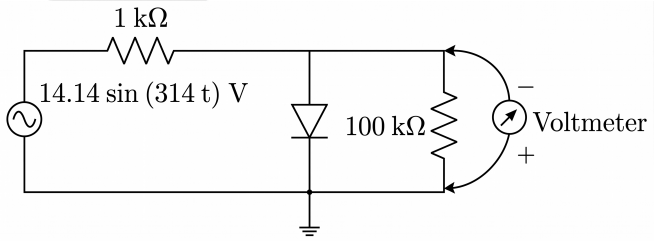


Fig. Q2.13.1

A. 4.46 **B.** 3.15 **C.** 2.23 **D.** 0

Q2.13.2 MCQ 2013 1M

Three moving iron type voltmeters are connected as shown below. Voltmeter readings are V, V_1 and V_2, as indicated. The correct relation among the voltmeter readings is

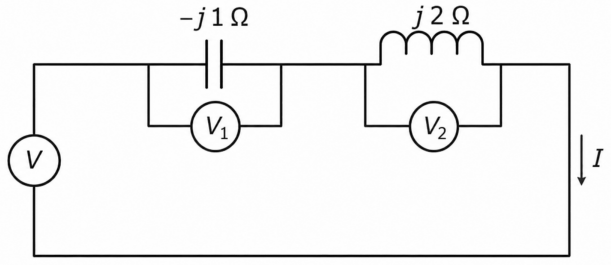


Fig. Q2.13.2

A. $V = \frac{V_1}{\sqrt{2}} + \frac{V_2}{\sqrt{2}}$ **B.** $V = V_1 + V_2$

C. $V = V_1 V_2$ **D.** $V = V_2 - V_1$

Q2.12.1 MCQ 2012 1M

For the circuit shown in the figure, the voltage and current expressions are $v(t) = E_1 \sin(\omega t) + E_3 \sin(3\omega t)$ and $i(t) = I_1 \sin(\omega t - \phi_1) + I_3 \sin(3\omega t - \phi_3) + I_5 \sin(5\omega t)$. The average power measured by the Wattmeter is

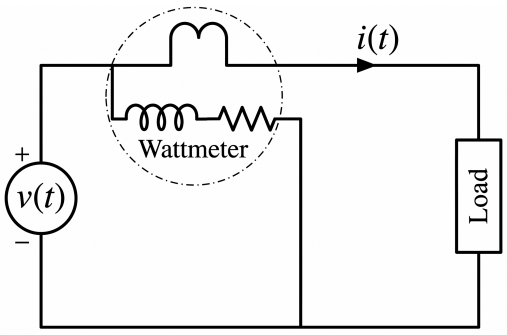


Fig. Q2.12.1

A. $\frac{1}{2}E_1 I_1 \cos\phi_1$

B. $\frac{1}{2}[E_1 I_1 \cos\phi_1 + E_1 I_3 \cos\phi_3 + E_1 I_5]$

C. $\frac{1}{2}[E_1 I_1 \cos\phi_1 + E_3 I_3 \cos\phi_3]$

D. $\frac{1}{2}[E_1 I_1 \cos\phi_1 + E_3 I_1 \cos\phi_1]$

Q2.12.2 MCQ 2012 2M

An analog voltmeter uses external multiplier settings. With a multiplier setting of 20 kΩ, it reads 440 V and with a multiplier setting of 80 kΩ, it reads 352 V. For a multiplier setting of 40 kΩ, the voltmeter reads

A. 371 V **B.** 383 V **C.** 394 V **D.** 406 V

Q2.12.3 MCQ 2012 1M

A periodic voltage waveform observed on an oscilloscope across a load is shown. A permanent magnet moving coil (PMMC) meter connected across the same load reads

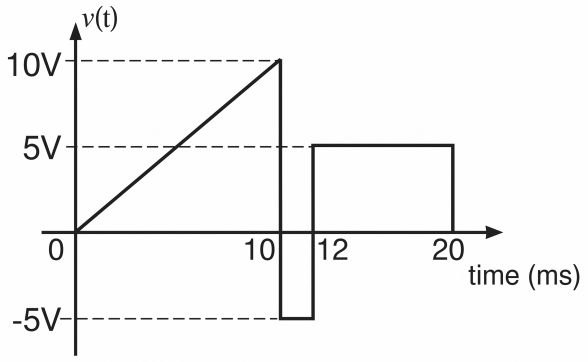


Fig. Q2.12.3

A. 4 V **B.** 5 V **C.** 8 V **D.** 10 V

Q2.11.1 MCQ 2011 1M

Consider the following statements:

(i) The compensating coil of a low power factor wattmeter compensates the effect of the impedance of the current coil.

(ii) The compensating coil of a low power factor wattmeter compensates the effect of the impedance of the voltage coil circuit.

A. (i) is true but (ii) is false

B. (i) is false but (ii) is true

C. both (i) and (ii) are true

D. both (i) and (ii) are false

Q2.10.1 MCQ 2010 1M

An ammeter has a current range of 0 – 5 A, and its internal resistance is 0.2 Ω. In order to change the range to 0 – 25 A, we need to add a resistance of

A. 0.8 Ω in series with the meter.

B. 1.0 Ω in series with the meter.

C. 0.04 Ω in parallel with the meter.

D. 0.05 Ω in parallel with the meter.

Q2.10.2 MCQ 2010 1M

A wattmeter is connected as shown in the figure. The wattmeter reads

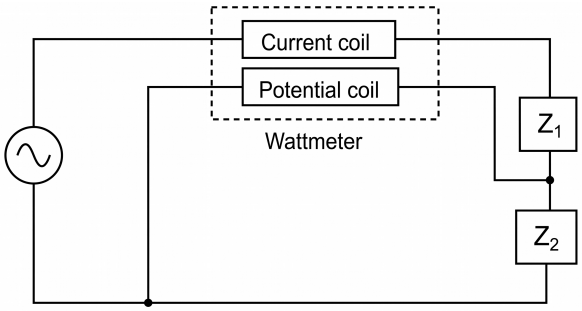


Fig. Q2.10.2

A. Zero always

B. Total power consumed by Z_1 and Z_2

C. Power consumed by Z_1

D. Power consumed by Z_2

Q2.09.1 MCQ 2009 2M

The figure shows a three-phase delta connected load supplied from a 400 V, 50 Hz, 3-phase balanced source. The pressure coil (PC) and current coil (CC) of a wattmeter are connected to the load as shown, with the coil polarities suitably selected to ensure a positive deflection. The wattmeter reading will be

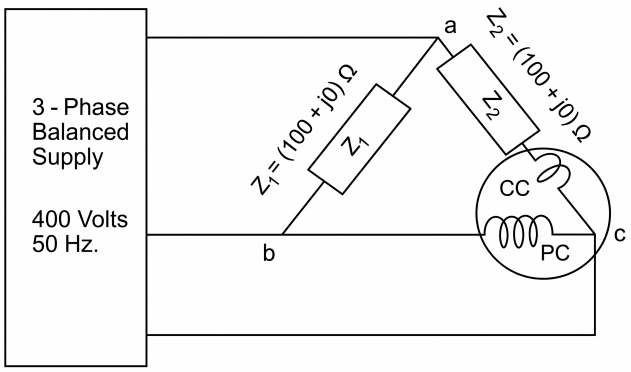


Fig. Q2.09.1

A. 0 **B.** 1600 Watt **C.** 800 Watt **D.** 400 Watt

Q2.09.2 MCQ 2009 1M

The pressure coil of a dynamometer type wattmeter is

A. highly inductive **B.** highly resistive

C. purely resistive **D.** purely inductive

Q2.06.1 MCQ 2006 2M

An energy meter connected to an immersion heater (resistive) operating on an AC 230 V, 50 Hz, AC single phase source reads 2.3 units (kWh) in 1 hour. The heater is removed from the supply and now connected to a 400 V peak-to-peak square wave source of 150 Hz. The power in kW dissipated by the heater will be

A. 3.478 **B.** 1.739 **C.** 1.540 **D.** 0.870

Q2.06.2 MCQ 2006 2M

A current of $-8 + 6\sqrt{2}(\sin \omega t + 30^\circ)$ A is passed through three meters. They are a centre zero PMMC meter, a true rms meter and a moving iron instrument. The respective readings (in A) will be

A. 8, 6, 10 **B.** 8, 6, 8 **C.** −8, 10, 10 **D.** −8, 2, 2

Q2.05.1 MCQ 2005 2M

A 1000 V DC supply has two single-core cables as its positive and negative leads; their insulation resistances to earth are 4 MΩ and 6 MΩ, respectively, as shown in the given figure. A voltmeter with resistance 50 kΩ is used to measure the insulation of the cable. When connected between the positive core and earth, then voltmeter reads

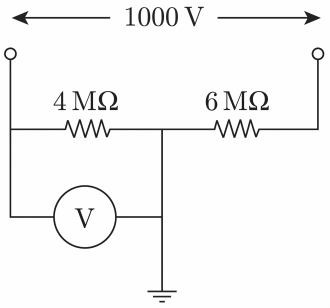


Fig. Q2.05.1

A. 8 V **B.** 16 V **C.** 24 V **D.** 40 V

Q2.04.1 MCQ 2004 2M

A single-phase load is connected between R and Y terminals of a 415 V, symmetrical, 3-phase, 4-wire system with phase sequence RYB. A wattmeter is connected in the system as shown in figure. The power factor of the load is 0.8 lagging. The wattmeter will read

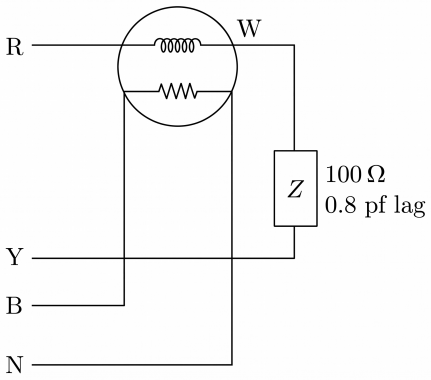


Fig. Q2.04.1

A. −795 W **B.** −597 W **C.** +597 W **D.** +795 W

Q2.04.2 MCQ 2004 2M

A moving-iron ammeter produces a full scale torque of 240 μNm with a deflection of 120° at a current of 10 A. The rate of change of self-inductance (μH/radian) of the instrument at full scale is

A. 2.0 μH/radian **B.** 4.8 μH/radian
C. 12.0 μH/radian **D.** 114.6 μH/radian

Q2.04.3 MCQ 2004 2M

A DC ampere hour meter is rated for 15 A, 250 V. The meter constant is 14.4 A seconds/rev. The meter constant at rated voltage may be expressed as

A. 3750 rev/kWh **B.** 3600 rev/kWh
C. 1000 rev/kWh **D.** 960 rev/kWh

Q2.04.4 MCQ 2004 2M

A moving coil of a meter has 100 turns, and a length and depth of 10 mm and 20 mm, respectively. It is positioned in a uniform radial flux density of 200 mT. The coil carries a current of 50 mA. The torque on the coil is

A. 200 μNm **B.** 100 μNm **C.** 2 μNm **D.** 1 μNm

Q2.04.5 MCQ 2004 1M

A galvanometer with a full scale current of 10 mA has a resistance of 1000 Ω. The multiplying power (the ratio of measured current to galvanometer current) of a 100 Ω shunt with this galvanometer is

A. 110 **B.** 100 **C.** 11 **D.** 10

Q2.04.6 MCQ 2004 1M

The circuit in the given figure is used to measure the power consumed by the load. The current coil and the voltage coil of the wattmeter have 0.02 Ω and 1000 Ω resistances, respectively. The measured power compared to the load power will be

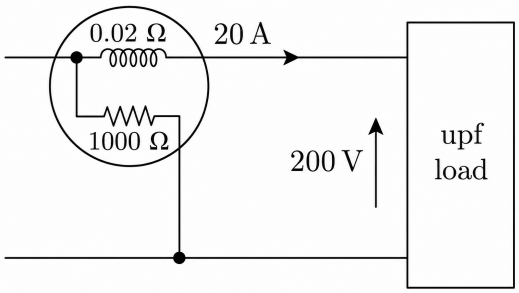


Fig. Q2.04.6

A. 0.4% less **B.** 0.2% less
C. 0.2% more **D.** 0.4% more

Q2.03.1 MCQ 2003 2M

The voltage-flux adjustment of a certain single-phase 220 V induction watt hour meter is altered so that the phase angle between the applied voltage and the flux due to it is 85° (instead of 90°). The errors introduced in the reading of this meter when the current is 5 A at power factors of unity and 0.5 lagging are, respectively,

A. 3.8 mW, 77.4 mW **B.** −3.8 mW, −77.4 mW
C. −4.2 mW, −85.1 mW **D.** 4.2 mW, 85.1 mW

Q2.03.2 MCQ 2003 2M

The inductance of a certain moving-iron ammeter is expressed as $L = \left(10 + 3\theta - \frac{\theta^2}{4}\right)$ μH, where θ is deflection in radians from the zero position. The control spring torque is 25×10^{-6} N-m/radian. The deflection of the pointer in radian when the meter carries a current of 5 A, is

A. 2.4 **B.** 2.0 **C.** 1.2 **D.** 1.0

Q2.03.3 MCQ 2003 2M

A wattmeter reads 400 W when its current coil is connected in the R phase and its pressure coil is connected between this phase and the neutral of a symmetrical three-phase system supplying a balanced star connected 0.8 pf inductive load. The phase sequence is RYB. What will be the reading of this wattmeter if its pressure coil alone is reconnected between the B and Y phases, all other connections remaining as before?

A. 400.0 **B.** 519.6 **C.** 300.0 **D.** 692.8

Q2.03.4 MCQ 2003 1M

The effect of stray magnetic fields on the actuating torque of a portable instrument is the maximum when the operating field of the instrument and the stray fields are

A. perpendicular **B.** parallel
C. inclined at 60° **D.** inclined at 30°

Q2.03.5 MCQ 2003 1M

A Manganin swamp resistance is connected in series with a moving coil ammeter consisting of a milli-ammeter and a suitable shunt in order to

A. minimise the effect of temperature variation.

B. obtain large deflecting torque.

C. reduce the size of the meter.

D. minimise the effect of stray magnetic fields.

Q2.03.6 MCQ 2003 2M

A rectifier type AC voltmeter consists of a series resistance R_s, an ideal full-wave rectifier bridge and a PMMC instrument as shown in the given figure. The internal resistance of the instrument is 100 Ω and a full scale deflection is produced by a DC current of 1 mA. The value of R_s required to obtain full scale deflection with an AC voltage of 100 V (rms) applied to the input terminals is

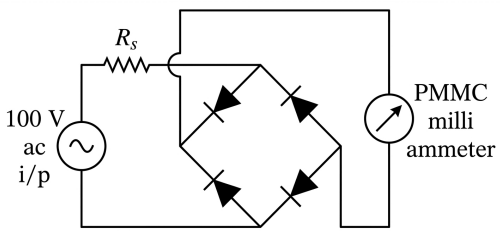


Fig. Q2.03.6

A. 63.56 Ω **B.** 69.93 Ω
C. 89.93 kΩ **D.** 141.3 kΩ

Q2.02.1 MCQ 2002 1M

The line-to-line voltage to the 3-phase, 50 Hz, AC circuit as shown in the following figure is 100 V rms. Assuming that the phase sequence is RYB, the wattmeters would read

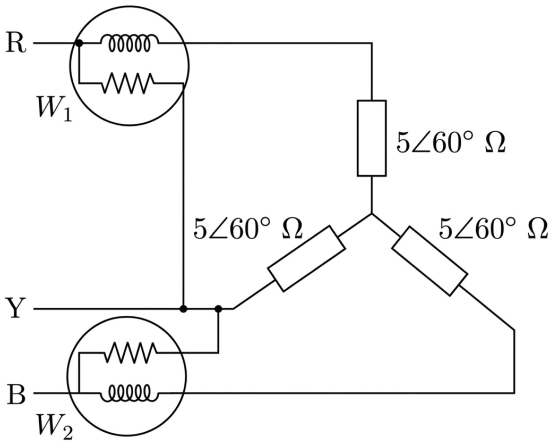


Fig. Q2.02.1

A. $W_1 = 886$ W and $W_2 = 886$ W

B. $W_1 = 500$ W and $W_2 = 500$ W

C. $W_1 = 0$ W and $W_2 = 1000$ W

D. $W_1 = 250$ W and $W_2 = 750$ W

Q2.01.1 MCQ 2001 1M

The minimum number of wattmeter(s) required to measure 3-phase, 3-wire balanced or unbalanced power is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q2.01.2 MCQ 2001 1M

If an energy meter disc makes 10 revolutions in 100 seconds when a load of 450 W is connected to it, the meter constant (in rev/kWh) is

A. 1000 **B.** 500 **C.** 1600 **D.** 800

Q2.00.1 MCQ 2000 2M

The two-wattmeter method is used to measure active power on a three phase, three wire system. If the phase voltage is unbalanced, then the power reading is:

A. affected by both negative sequence and zero sequence voltages

B. affected by negative sequence voltages but not by zero sequence voltages

C. affected by zero sequence voltage but not by negative sequence voltages

D. not affected by negative or zero sequence voltages

Q2.00.2 MCQ 2000 1M

An analog electronic circuit that measures *rms* value of the input voltage by averaging the square of the instantaneous voltage level, responds slowly to changes in the input signal due to

A. the "square" function built into the circuit

B. the "square-root" function built into the circuit

C. the averaging function built into the circuit

D. none of the above

Q2.00.3 NAT 2000 5M

A $3-\phi$ load operates with balanced voltages applied to its terminals, and draws balanced currents. The potential coil of a moving coil wattmeter is connected from R to Y terminals of the load. The current coil of the meter is connected in series with phase B. By appropriate derivation, show that the quantity indicated by this wattmeter is proportional to the reactive power drawn by the load.

Q2.99.1 MCQ 1999 1M

Electro dynamic type watt meters have large errors while measuring power in a.c circuits at low power factor conditions, since the voltage across and the current through the

A. current coil are not in phase

B. current coil are not in quadrature

C. pressure coil are not in phase

D. pressure coil are not in quadrature

Q2.99.2 NAT 1999 5M

For the given circuit the internal resistance of the ammeters as well as that of the current coil of the wattmeter are zero, and the voltmeters have a very large figure of merit. A_1 is a 6 A full scale moving Iron type meter. A_2 is a commercial full wave rectifier type meter of full scale 5 A. V_1 is a 500 V meter is an peak response type. V_2 is a 300 V PMMC type meter and W is an electro dynamometer type, 5 A, 230 V wattmeter. Determine the readings of A_1, A_2, V_1, V_2, and W.

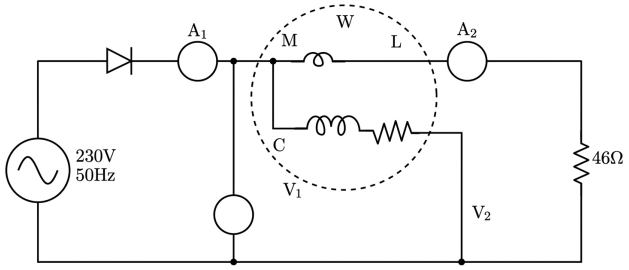


Fig. Q2.99.2

Q2.98.1 MCQ 1998 1M

A d.c. voltmeter has a sensitivity of 1000 Ω/V. When it measures half full scale in 100 V range, the current through the voltmeter is

A. 100 mA **B.** 1 mA **C.** 0.5 mA **D.** 50 mA

Q2.98.2 NAT 1998 5M

A symmetrical 400 V, $3-\phi$ supply is connected to the network shown in fig. The phase sequence is RYB. Find the reading on the wattmeter. $R_1 = 30\,\Omega$, $X_1 = 50\,\Omega$, and $X_2 = 40\,\Omega$

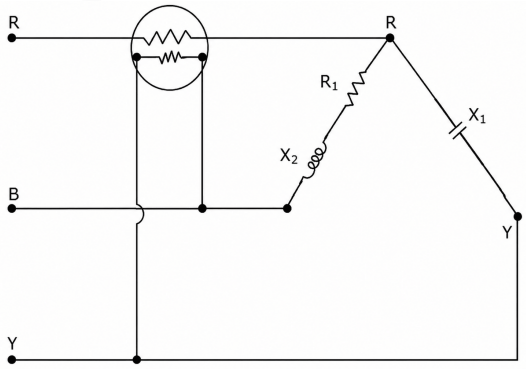


Fig. Q2.98.2

Q2.97.1 MCQ 1997 1M

A dynamometer type wattmeter responds to the

A. average value of active power

B. average value of reactive power

C. peak value of active power

D. peak value of reactive power

Q2.96.1 MCQ 1996 1M

An advantage of a permanent magnet moving coil instrument is that it is

A. free from friction error

B. has high (torque/weight of the moving parts) ratio

C. has low (torque/weight of the moving parts) ratio

D. can be used on both a.c. and d.c.

Q2.96.2 MCQ 1996 1M

The scale of a voltmeter is uniform. Its type is

A. moving iron

B. induction

C. moving coil permanent magnet

D. moving coil dynamometer

Q2.96.3 MCQ 1996 1M

The moving coil in a dynamometer wattmeter is connected

A. in series with the fixed coil

B. across the supply

C. in series with the load

D. across the load

Q2.96.4 MCQ 1996 1M

For a given frequency, the deflecting torque of an induction ammeter is directly proportional to

A. current^2 **B.** current^3
C. $\sqrt{\text{current}}$ **D.** current

Q2.95.1 NAT 1995 1M

Two 100 V full scale PMMC type dc voltmeter having figure of merits (FOM) of 10 kΩ/V and 20 kΩ/V are connected in series. The series combination can be used to measure a maximum DC voltage of _____ V.

Q2.94.1 MCQ 1994 1M

A 0 – 10 mA PMMC ammeter reads 4 mA in a circuit. Its bottom control spring snaps suddenly. The meter will now read nearly

A. 10 mA **B.** 8 mA **C.** 2 mA **D.** zero

Q2.93.1 MCQ 1993 1M

A metal strain gauge has factor of two. Its nominal resistance is 120 Ω. It undergoes a strain of 10^{-5}, the value of change of resistance in response to the strain is

A. 240 Ω **B.** 2×10^{-5} Ω
C. 2.4×10^{-3} Ω **D.** 1.2×10^{-3} Ω

Q2.92.1 MCQ 1992 1M

An unshielded moving iron voltmeter is used to measure the voltage in an ac circuit. If a stray DC magnetic field having a component along the axis of the meter coil appears, the meter reading would be

A. unaffected

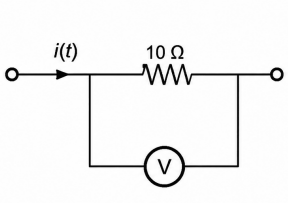
B. decreased

C. increased

D. either decreased or increased depending on the direction of the DC field

Q2.91.1 NAT 1991 1M

The current $i(t)$ passing through 10 Ω resistor as shown in the Fig. (i) as a waveform as shown in Fig. (ii). Then the reading of the DC voltmeter connected across 10 Ω resistor is



D.C. Voltmeter
Fig. (i)

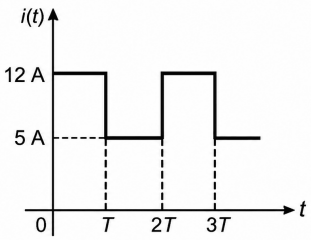


Fig. (ii)

Fig. Q2.91.1

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q2.24.1 692 to 693	Q2.23.1 C	Q2.20.1 1.700 to 1.750
Q2.19.1 D	Q2.18.1 D	Q2.18.2 A
Q2.17.1 D	Q2.16.1 0.30 to 0.33	Q2.16.2 1.9 to 2.1
Q2.15.1 0.22 to 0.23	Q2.15.2 C	Q2.15.3 A
Q2.15.4 0.14 to 0.16	Q2.14.1 2 to 2	Q2.14.2 0.78 to 0.82
Q2.14.3 C	Q2.14.4 A	Q2.14.5 9.9 to 10.3
Q2.14.6 56 to 59	Q2.14.7 3 to 3.4	Q2.13.1 A
Q2.13.2 D	Q2.12.1 C	Q2.12.2 D
Q2.12.3 A	Q2.11.1 B	Q2.10.1 D
Q2.10.2 D	Q2.09.1 C	Q2.09.2 B
Q2.06.1 B	Q2.06.2 C	Q2.05.1 A
Q2.04.1 B	Q2.04.2 B	Q2.04.3 C
Q2.04.4 A	Q2.04.5 C	Q2.04.6 C
Q2.03.1 C	Q2.03.2 C	Q2.03.3 B
Q2.03.4 B	Q2.03.5 A	Q2.03.6 C
Q2.02.1 C	Q2.01.1 B	Q2.01.2 D
Q2.00.1 D	Q2.00.2 C	Q2.00.3 *
Q2.99.1 C	Q2.99.2 *	Q2.98.1 C
Q2.98.2 -404.8	Q2.97.1 A	Q2.96.1 B
Q2.96.2 C	Q2.96.3 B	Q2.96.4 A
Q2.95.1 150 to 150	Q2.94.1 D	Q2.93.1 C
Q2.92.1 D	Q2.91.1 85 to 85	

UNIT

III

DC And AC Bridges

Topics

1. Wheatstone Bridge
2. AC Bridges: Maxwell, Hay, Schering, DeSauty, Wien Bridge for measuring Inductance and Capacitance

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 13	**MCQ:** 9	**MSQ:** 0	**NAT:** 4
1M: 5	**2M:** 8	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2021)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 NAT 2026 2M

The resistance values of the Wheatstone bridge shown are: $P = 2000\,\Omega$, $Q = 500\,\Omega$, $R = 3000\,\Omega$. The battery voltage $E = 50\,\text{V}$. The battery has an internal resistance of $1\,\Omega$ and the Galvanometer (G) has a resistance of $50\,\Omega$.

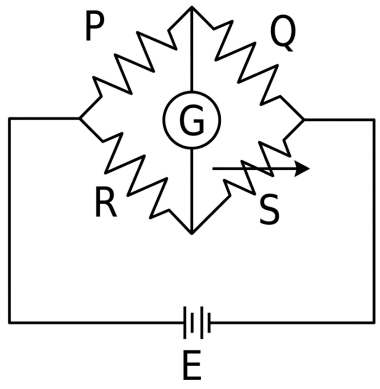


Fig. Q3.26.1

The value of the resistance S for balanced condition is _____ Ω (Answer in integer).

Q3.25.1 NAT 2025 2M

In the Wheatstone bridge shown below, the sensitivity of the bridge in terms of change in balancing voltage E for unit change in the resistance R, in mV/Ω, is _____ (round off to two decimal places).

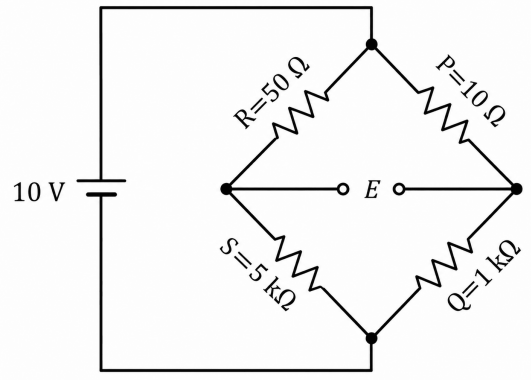


Fig. Q3.25.1

Q3.21.1 MCQ 2021 1M

Inductance is measured by,

A. Schering bridge
B. Maxwell bridge
C. Kelvin bridge
D. Wien bridge

Q3.15.1 MCQ 2015 2M

An unbalanced DC Wheatstone bridge is shown in the figure. At what value of p will the magnitude of V_0 be maximum?

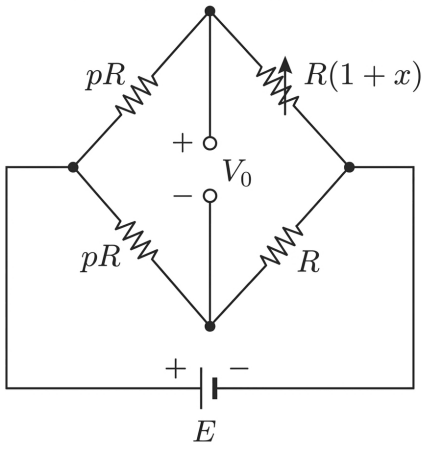


Fig. Q3.15.1

A. $\sqrt{(1+x)}$
B. $(1+x)$
C. $\frac{1}{\sqrt{(1+x)}}$
D. $\sqrt{(1-x)}$

Q3.14.1 NAT 2014 2M

In the bridge circuit shown, the capacitors are loss free. At balance, the value of capacitance C_1 in microfarad is _____.

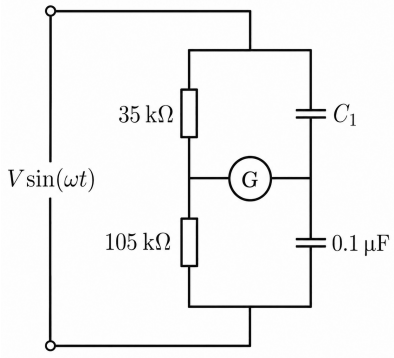


Fig. Q3.14.1

Q3.14.2 NAT 2014 2M

The reading of the voltmeter (rms) in volts, for the circuit shown in the figure is _____

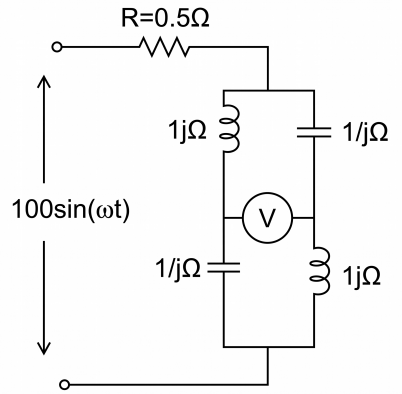


Fig. Q3.14.2

Q3.13.1 MCQ 2013 2M

A strain gauge forms one arm of the bridge shown in the following figure and has a nominal resistance without any load as $R_s = 300\,\Omega$. Other bridge resistances are $R_1 = R_2 = R_3 = 300\,\Omega$. The maximum permissible current through the strain gauge is 20 mA. During certain measurement when the bridge is excited by maximum permissible voltage and the strain gauge resistance is increased by 1% over the nominal value, the output voltage V_o in mV is

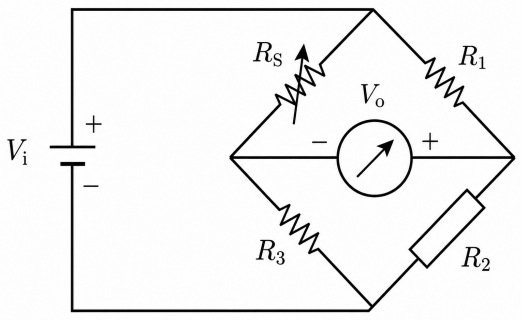


Fig. Q3.13.1

A. 56.02 **B.** 40.83
C. 29.85 **D.** 10.02

Q3.12.1 MCQ 2012 1M

The bridge method commonly used for finding mutual inductance is

A. Heaviside Campbell bridge

B. Schering bridge

C. De Sauty bridge

D. Wein bridge

Q3.11.1 MCQ 2011 1M

The bridge circuit shown in the following figure is used for the measurement of an unknown element Z_x. The bridge circuit is best suited when Z_x is a

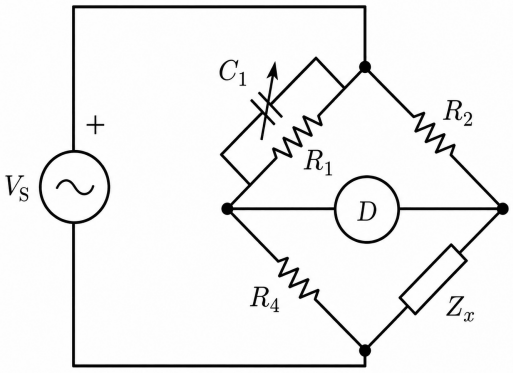


Fig. Q3.11.1

A. low resistance **B.** high resistance
C. low Q inductor **D.** lossy capacitor

Q3.10.1 MCQ 2010 2M

The Maxwell's bridge shown in the figure is at balance. The parameters of the inductive coil are.

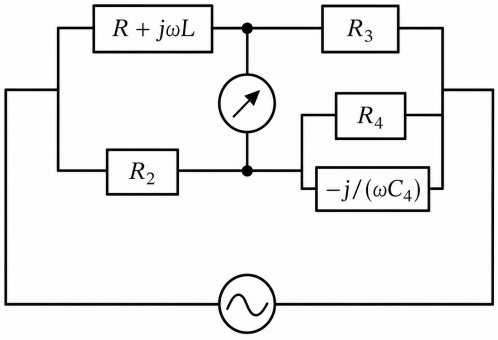


Fig. Q3.10.1

A. $R = R_2R_3/R_4, L = C_4R_2R_3$

B. $L = R_2R_3/R_4, R = C_4R_2R_3$

C. $R = R_4/R_2R_3, L = C_4R_2R_3$

D. $L = R_4/R_2R_3, R = 1/(C_4R_2R_3)$

Q3.08.1 MCQ 2008 1M

The ac bridge shown in the figure is used to measure the impedance Z. If the bridge is balanced for oscillator frequency $f = 2\,\text{kHz}$, then the impedance Z will be

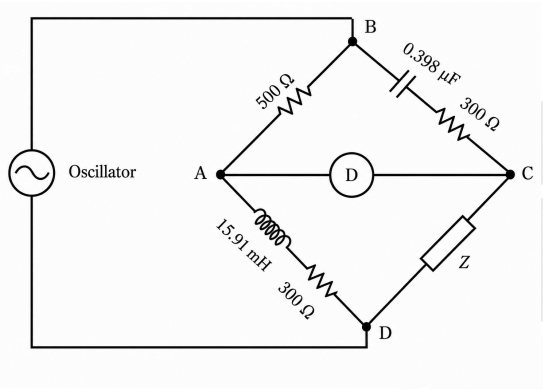


Fig. Q3.08.1

A. $(260 + j0)\ \Omega$

B. $(0 + j200)\ \Omega$

C. $(260 - j200)\ \Omega$

D. $(260 + j200)\ \Omega$

Q3.07.1 MCQ 2007 1M

A bridge circuit is shown in the figure below. Which one of the sequence given below is most suitable for balancing the bridge?

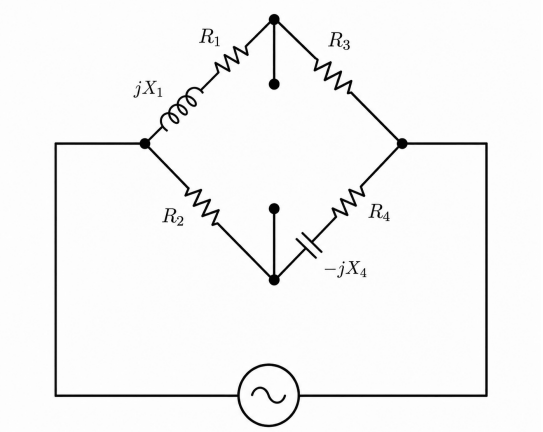


Fig. Q3.07.1

A. First adjust R_4, and then adjust R_1

B. First adjust R_2, and then adjust R_3

C. First adjust R_2, and then adjust R_4

D. First adjust R_4, and then adjust R_2

Q3.03.1 MCQ 2003 2M

The items in Group I represent the various types of measurements to be made with a reasonable accuracy using a suitable bridge. The items in Group II represent the various bridges available for this purpose. Select the correct choice of the item in Group II for the corresponding item in Group I from the following.

Group I	Group II
P. Resistance in the milliohm range Q. Low values of capacitance R. Comparison of resistances which are nearly equal S. Inductance of a coil with a large time constant	1. Wheatstone bridge 2. Kelvin double bridge 3. Schering bridge 4. Wien's bridge 5. Hay's bridge 6. Carey-Foster bridge

A. P – 2 Q – 3 R – 6 S – 5

B. P – 2 Q – 6 R – 4 S – 5

C. P – 2 Q – 3 R – 5 S – 4

D. P – 1 Q – 3 R – 2 S – 6

Answer Key

Q3.26.1 750 to 750	Q3.25.1 -2.00 to -1.94 OR 1.94 to 2.00	Q3.21.1 B
Q3.15.1 A	Q3.14.1 0.3 to 0.3	Q3.14.2 141.42
Q3.13.1 C	Q3.12.1 A	Q3.11.1 C
Q3.10.1 A	Q3.08.1 A	Q3.07.1 A
Q3.03.1 A		

UNIT

IV

Measurement of Resistance

Topics

1. Kelvin Double Bridge
2. V-A, A-V method
3. Loss of Charge Method
4. Ohmmeter, Potentiometer

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 5	**MCQ:** 5	**MSQ:** 0	**NAT:** 0
1M: 4	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~0M	**Range:** 0M(–)–0M(–)

**See preface for how Avg Weightage and Range are calculated.*

Q4.02.1 MCQ 2002 1M

Kelvin double bridge is best suited for the measurement of:

A. Resistances of very low value

B. Low value capacitance

C. Resistance of very high value

D. High value capacitance

Q4.97.1 MCQ 1997 1M

A potentiometer is basically

A. Deflection type instrument

B. Null type instrument

C. Deflection as well as null

D. A digital instrument

Q4.97.2 MCQ 1997 1M

A transfer instrument employed in the standardization of a polar type $a.c$ potentiometer is _____

A. an electro static instrument

B. a thermal instrument

C. a dynamo meter instrument

D. a moving coil instrument

Q4.94.1 MCQ 1994 2M

A d.c. potentiometer is designed to measure up to about $2\,V$ with a slide wire of 800 mm. A standard cell of emf $1.18\,V$ obtains balance at 600 mm. A test cell is seen to obtain balance at 680 mm. The emf of the test cell is

A. $1.00\,V$ **B.** $1.34\,V$ **C.** $1.50\,V$ **D.** $1.70\,V$

Q4.92.1 MCQ 1992 1M

In $d.c.$ potentiometer measurements, a second reading is often taken after reversing the polarities of the $s.c.$ supply and the unknown voltage, and the average of the two readings is taken. This is, with a view to eliminate the effects of

A. Ripples in the $d.c.$ supply

B. Stray magnetic fields

C. Stray thermal emf's

D. Erroneous standardization

Answer Key

Q4.02.1 A	Q4.97.1 B	Q4.97.2 B
Q4.94.1 B	Q4.92.1 C	

UNIT

V

Electronic Measurement and Instrument Transformer

Topics

1. CRO
2. Current Transformer CT, Potential Transformer PT
3. Digital Voltmeter - DVM
4. Q-Meter

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 34	**MCQ:** 32	**MSQ:** 0	**NAT:** 2
1M: 19	**2M:** 14	**Descriptive:** 1	**Avg Weightage** ~2M	**Range:** 1M(2024)–3M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 2M

In the circuit shown, the phase currents are:

$$I_A = 572.812 + j50.115 \text{ A}$$
$$I_B = -254.525 - j459.175 \text{ A}$$
$$I_C = -207.083 + j444.091 \text{ A}$$

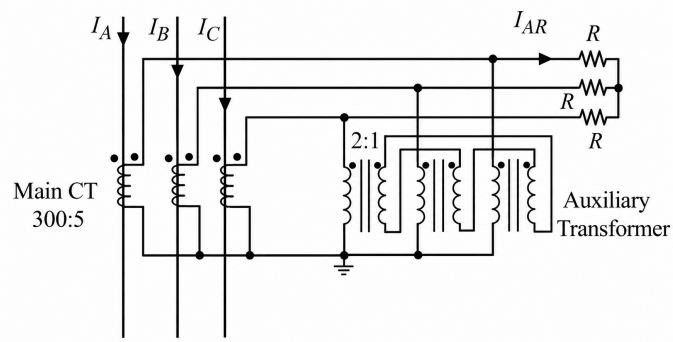


Fig. Q5.26.1

Given that the CTs are ideal with no saturation, and the turns ratio of the Main CT is 300 : 5 and that of the Auxiliary Transformer (YnΔ) is 2 : 1 on every phase, the value of I_{AR}, rounded off to three decimal places, is:

A. 0 A **B.** $0.653\angle 17.556°$ A
C. $537.240\angle 4.105°$ A **D.** $8.954\angle 4.105°$ A

Q5.24.1 MCQ 2024 1M

The table lists two instrument transformers and their features:

Instrument Transformers	Features
	(P) Primary is connected in parallel to the grid
(X) Current Transformer (*CT*)	(Q) Open circuited secondary is not desirable
(Y) Potential Transformer (*PT*)	(R) Primary current is the line current
	(S) Secondary burden affects the primary current

The correct matching of the two column is

A. X matches with P and Q; Y matches with R and S.

B. X matches with P and R; Y matches with Q and S.

C. X matches with Q and R; Y matches with P and S.

D. X matches with Q and S; Y matches with P and R.

Q5.17.1 MCQ 2017 1M

A stationary closed Lissajous pattern on an oscilloscope has 3 horizontal tangencies and 2 vertical tangencies for a horizontal input with, frequency 3 kHz. The frequency of the vertical input is

A. 1.5 kHz **B.** 2 kHz **C.** 3 kHz **D.** 4.5 kHz

Q5.17.2 MCQ 2017 1M

The slope and level detector circuit in a CRO has a delay of 100 ns. The start-stop sweep generator has a response time of 50 ns. In order to display correctly, a delay line of

A. 150 ns has to be inserted into the y-channel

B. 150 ns has to be inserted into the x-channel

C. 150 ns has to be inserted into both x and y channels

D. 100 ns has to be inserted into both x and y channels

Q5.14.1 MCQ 2014 1M

The two signals $S1$ and $S2$, shown in figure, are applied to Y and X deflection plates of an oscilloscope. The waveform displayed on the screen is

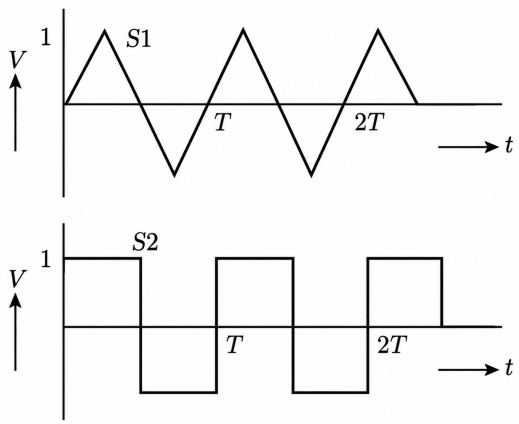
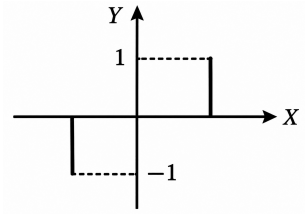
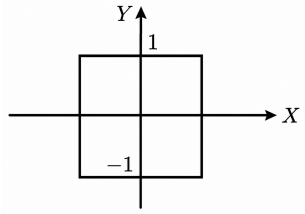


Fig. Q5.14.1

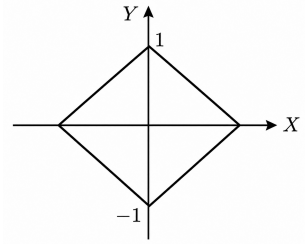
A.



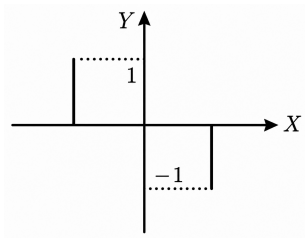
B.



C.



D.



Q5.14.2 MCQ 2014 1M

In an oscilloscope screen, linear sweep is applied at the

A. vertical axis.

B. horizontal axis.

C. origin.

D. both horizontal and vertical axis.

Q5.11.1 MCQ 2011 2M

A $4\frac{1}{2}$ digit DMM has the error specification as: 0.2% of reading + 10 counts. If a DC voltage of 100 V is read on its 200 V full scale, the maximum error that can be expected in the reading is

A. ±0.1% **B.** ±0.2% **C.** ±0.3% **D.** ±0.4%

Q5.11.2 MCQ 2011 1M

A dual trace oscilloscope is set to operate in the ALTernate mode. The control input of the multiplexer used in the y-circuit is fed with a signal having a frequency equal to

A. the highest frequency that the multiplexer can operate properly

B. twice the frequency of the time base (sweep) oscillator

C. the frequency of the time base (sweep) oscillator

D. half the frequency of the time base (sweep) oscillator

Q5.09.1 MCQ 2009 2M

An average-reading digital multimeter reads 10 V when fed with a triangular wave, symmetric about the time-axis. For the same input an rms reading meter will read

A. $\frac{20}{\sqrt{3}}$ **B.** $\frac{10}{\sqrt{30}}$ **C.** $20\sqrt{30}$ **D.** $10\sqrt{3}$

Q5.09.2 MCQ 2009 1M

The two inputs of a CRO are fed with two stationary periodic signals. In the X-Y mode, the screen shows a figure which changes from ellipse to circle and back to ellipse with its major axis changing orientation slowly and repeatedly. The following inference can be made from this.

A. The signals are not sinusoidal

B. The amplitudes of the signals are very close but not equal

C. The signals are sinusoidal with their frequencies very close but not equal

D. There is a constant but small phase difference between the signals

Q5.08.1 MCQ 2008 2M

Two sinusoidal signals $p(\omega_1 t) = A \sin \omega_1 t$ and $q(\omega_2 t)$ are applied to X and Y inputs of a dual channel CRO. The Lissajous figure displayed on the screen is shown in the following figure.

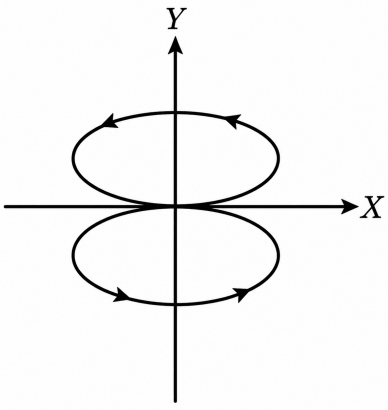


Fig. Q5.08.1

The signal $q(\omega_2 t)$ will be represented as

A. $q(\omega_2 t) = A \sin \omega_2 t,\ \omega_2 = 2\omega_1$

B. $q(\omega_2 t) = A \sin \omega_2 t,\ \omega_2 = \frac{\omega_1}{2}$

C. $q(\omega_2 t) = A \cos \omega_2 t,\ \omega_2 = 2\omega_1$

D. $q(\omega_2 t) = A \cos \omega_2 t,\ \omega_2 = \frac{\omega_1}{2}$

Q5.07.1 MCQ 2007 1M

The probes of a non-isolated, two-channel oscilloscope are clipped to points A, B and C in the circuit of the given figure. V_{in} is a square wave of a suitable low frequency. The display on Ch_1 and Ch_2 are as shown on the right. Then the "signal" and "ground" probes S_1, G_1 and S_2, G_2 of Ch_1 and Ch_2, respectively, are connected to points

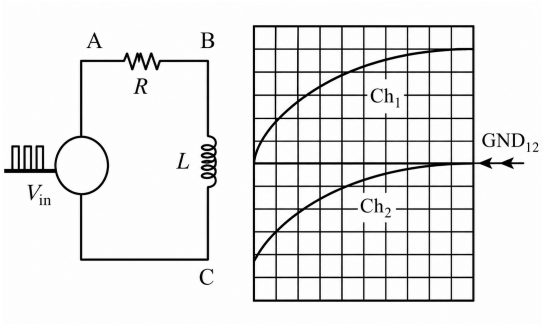


Fig. Q5.07.1

A. A, B, C, A **B.** A, B, C, B
C. C, B, A, B **D.** B, A, B, C

Q5.06.1 MCQ 2006 2M

A 200/1 current transformer (CT) is wound with 200 turns on the secondary on a toroidal core. When it carries a current of 160 A on the primary, the ratio and phase errors of the CT are found to be −0.5% and 30 minutes, respectively. If the number of secondary turns is reduced by 1, the new ratio error (%) and phase error (min) will be, respectively,

A. 0.0, 30 **B.** −0.5, 35
C. −1.0, 30 **D.** −1.0, 25

Q5.06.2 MCQ 2006 1M

A sampling wattmeter (that computes power from simultaneously sampled values of voltage and current) is used to measure the average power of a load. The peak to peak voltage of the square wave is 10 V and the current is a triangular wave of 5 A p-p as shown in the given figure. The period is 20 ms. The reading in W will be

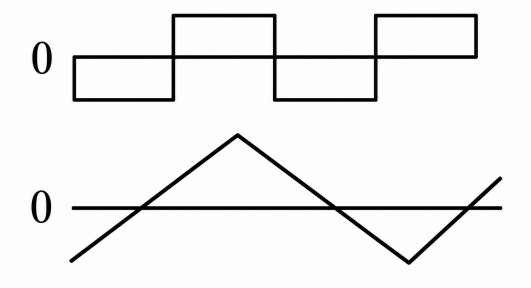


Fig. Q5.06.2

A. 0 W **B.** 25 W **C.** 50 W **D.** 100 W

Q5.06.3 MCQ 2006 1M

The time/div and voltage/div axes of an oscilloscope have been erased. A student connects a 1 kHz, 5 V p-p square wave calibration pulse to channel-1 of the scope and observes the screen to be as shown in the upper trace of the figure. An unknown signal is connected to channel-2 (lower trace) of the scope. If the time/div and V/div on both channels are the same, the amplitude (p-p) and period of the unknown signal are respectively

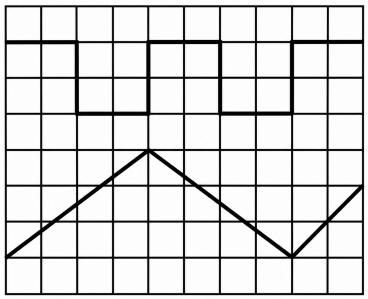


Fig. Q5.06.3

A. 5 V, 1 ms **B.** 5 V, 2 ms
C. 7.5 V, 2 ms **D.** 10 V, 1 ms

Q5.05.1 MCQ 2005 2M

The simultaneous application of signals $x(t)$ and $y(t)$ to the horizontal and vertical plates, respectively, of an oscilloscope, produces a vertical figure-of-8 display. If P and Q are constants, and $x(t) = P\sin(4t + 30°)$, then $y(t)$ is equal to

A. $Q\sin(4t - 30°)$ **B.** $Q\sin(2t + 15°)$
C. $Q\sin(8t + 60°)$ **D.** $Q\sin(4t + 30°)$

Q5.05.2 MCQ 2005 1M

The Q-meter works on the principle of

A. mutual inductance **B.** self inductance
C. series resonance **D.** parallel resonance

Common Data Questions

Q5.04.1 MCQ 2004 2M

A 50 Hz, bar primary CT has a secondary with 500 turns. The secondary supplies 5 A current into a purely resistive burden of 1 Ω. The magnetising ampere-turns is 200. The phase angle between the primary and secondary current is

A. 4.6° **B.** 85.4° **C.** 94.6° **D.** 175.4°

Q5.04.2 MCQ 2004 2M

The core flux in the CT of Question Q5.04.01 under the given operating condition is

A. 0 **B.** 45.0 μWb **C.** 22.5 μWb **D.** 100.0 μWb

Q5.04.3 MCQ 2004 2M

A CRO probe has an impedance of 500 kΩ in parallel with a capacitance of 10 pF. The probe is used to measure the voltage between P and Q as shown in the following figure. The source is 10 V RMS at 100 kHz; the 100 V source label in the redrawn figure is a misprint. The measured voltage will be

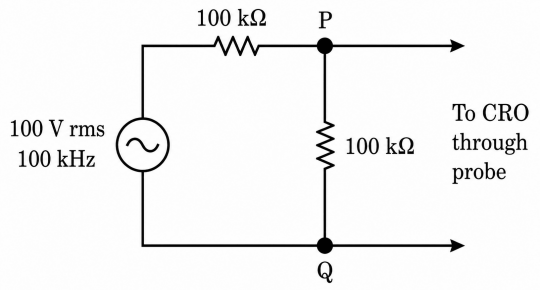


Fig. Q5.04.3

A. 3.53 V **B.** 4.37 V **C.** 4.54 V **D.** 5.00 V

Q5.03.1 MCQ 2003 1M

Group-II represents the figures obtained on a CRO screen when the voltage signals $V_x = V_{xm} \sin \omega t$ and $V_y = V_{ym} \sin(\omega t + \Phi)$ are given to its X and Y plates respectively and Φ is changed. Choose the correct value of Φ from Group-I to match with the corresponding figure of Group-II

Group 1

P. $\Phi = 0$

Q. $\Phi = \frac{\pi}{2}$

R. $\pi < \Phi < \frac{3\pi}{2}$

S. $\Phi = \frac{3\pi}{2}$

Group 2

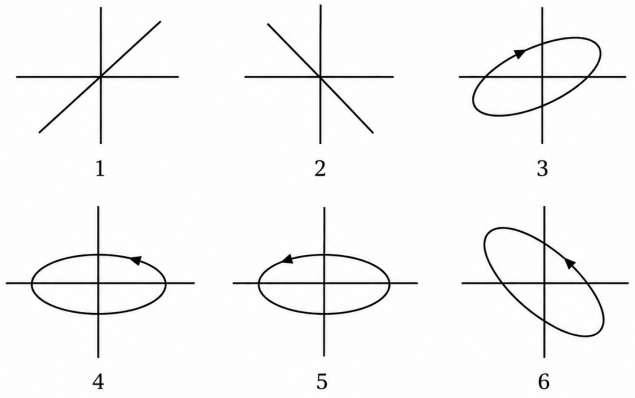


Fig. Q5.03.1

The correct matching of the two columns is

A. P-1, Q-3, R-6, S-5

B. P-2, Q-6, R-4, S-5

C. P-2, Q-3, R-5, S-4

D. P-1, Q-5, R-6, S-4

Q5.03.2 MCQ 2003 2M

A 500 A/5 A, 50 Hz current transformer has a bar primary. The secondary burden is a pure resistance of 1 Ω and it draws a current of 5 A. If the magnetic core requires 250 AT for magnetisation, the percentage ratio error is

A. 10.56 **B.** −10.56 **C.** 11.80 **D.** −11.80

Q5.03.3 MCQ 2003 1M

A reading of 120 is obtained when standard inductor was connected in the circuit of a Q-meter and the variable capacitor is adjusted to value of 300 pF. A lossless capacitor of unknown value C_x is then connected in parallel with the variable capacitor and the same reading was obtained when the variable capacitor is readjusted to a value of 200 pF. The value of C_x in pF is

A. 100 **B.** 200 **C.** 300 **D.** 500

Q5.02.1 MCQ 2002 1M

Two in-phase, 50 Hz sinusoidal waveform of unit amplitude are fed into channel 1 and channel 2 respectively of an oscilloscope. Assuming that the voltage scale, time scale and other settings are exactly the same for both the channels, what would be observed if the oscilloscope is operated in X-Y mode?

A. A circle of unit radius

B. An ellipse

C. A parabola

D. A straight line inclined at 45° with respect to the x-axis

Q5.00.1 MCQ 2000 1M

Instrument transformers are known to introduce magnitude and phase errors in measurements. These are primarily due to

A. improper connections on the primary side

B. measurement errors inherent in the meter connected to the transformer secondary

C. open and short circuit parameters of the instrument transformers

D. None of the above

Q5.99.1 MCQ 1999 2M

For a dual ADC type $3\frac{1}{2}$ digit DVM, the reference voltage is 100 mV and the first integration time is set to 300 ms. For some input voltage, the "deintegration "period is 370.2 ms. The DVM will indicate:

A. 123.4 **B.** 199.9
C. 100.0 **D.** 1.414

Q5.98.1 MCQ 1998 2M

The figure shows the input attenuator of a multimeter. The meter reads full-scale with 12 V at M with the range switch at position q. Obtain the voltage at M for full-scale deflection when the range switch is at position s.

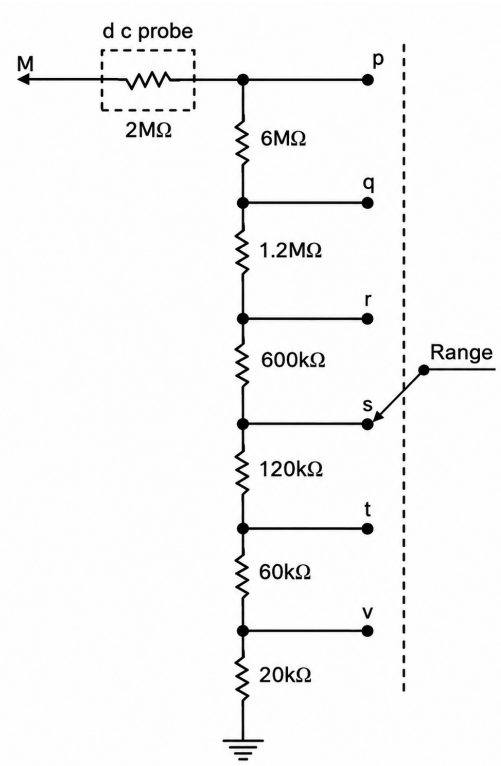


Fig. Q5.98.1

A. 1 V **B.** 150 V
C. 120 V **D.** 147 V

Q5.98.2 MCQ 1998 2M

Fig. shows the electrostatic vertical deflection system of CRT. Given that V_A is the accelerating voltage, the deflection sensitivity (deflection/volt) is proportional to

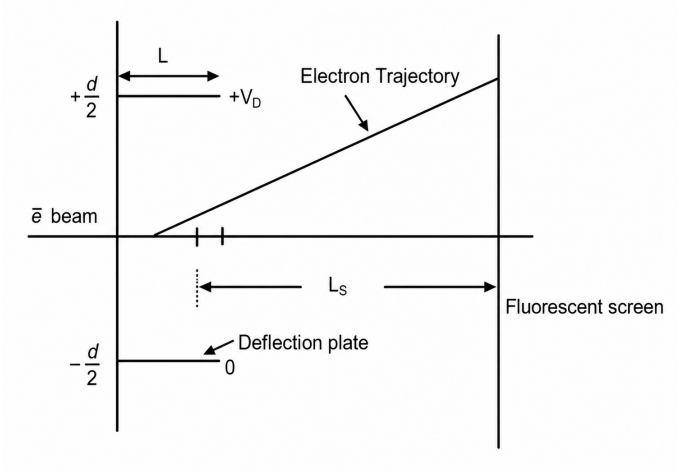


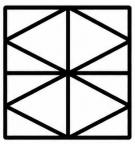
Fig. Q5.98.2

A. $\frac{LL_S}{dV_A}$ **B.** $\frac{LL_SV_D}{dV_A}$ **C.** $\frac{LL_S}{d}V_A$ **D.** $\frac{DL_S}{LV_A}$

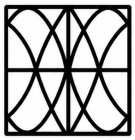
Q5.96.1 NAT 1996 3M

In an oscilloscope the input to the horizontal plates is a 100 Hz voltage signal. Lissajous patterns (A), (B) and (C) will be generated when different frequency voltage signals are applied to vertical plates. Match each lissajous pattern to its corresponding frequency.

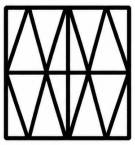
(a)



(b)



(c)



(P) $fy = 50$
(Q) $fy = 66.66$
(F) $fy = 125$
(S) $fy = 150$
(T) $fy = 200$
(U) $fy = 300$

Fig. Q5.96.1

Q5.95.1 MCQ 1995 1M

A certain oscilloscope with 4 cm × 4 cm screen has its own sweep output fed to its input. If the x and y sensitivities are same, the oscilloscope will display a

A. triangular wave **B.** diagonal line
C. sine wave **D.** circle

Q5.95.2 MCQ 1995 1M

A $3\frac{1}{2}$ digit, 2 V full scale dual slope ADC has its integration time set to 300 ms. If the input to the ADC is $(1 + 1 \sin 314t)$ V, then the ADC output will be

A. 1.000 **B.** 1.999 **C.** 1.414 **D.** 1.500

Q5.94.1 MCQ 1994 1M

A lissajous pattern, as shown in figure below, is observed on the screen of a CRO when voltages of frequencies f_x and f_y are applied to the x and y plates respectively. $f_x : f_y$ is then equal to

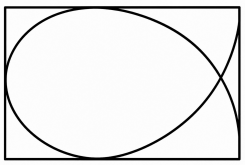


Fig. Q5.94.1

A. 3 : 2 **B.** 1 : 2 **C.** 2 : 3 **D.** 2 : 1

Q5.93.1 MCQ 1993 1M

A CRO screen has ten divisions on the horizontal scale. If a voltage signal $5 \sin(314t + 45°)$ is examined with a line base setting of 5 msec/div, the number of cycles of signal displayed on the screen will be

A. 0.5 cycles **B.** 2.5 cycles
C. 5 cycles **D.** 10 cycles

Q5.92.1 NAT 1992 2M

In a dual slope integrating type digital voltmeter the first integration is carried out for 10 periods of the supply frequency of 50 Hz. If the reference voltage used is 2 V, the total conversion time for an input 1 V is _____ sec

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q5.26.1 D	Q5.24.1 C	Q5.17.1 D
Q5.17.2 A	Q5.14.1 A	Q5.14.2 B
Q5.11.1 C	Q5.11.2 C	Q5.09.1 A
Q5.09.2 D	Q5.08.1 D	Q5.07.1 B
Q5.06.1 A	Q5.06.2 A	Q5.06.3 C
Q5.05.1 B	Q5.05.2 C	Q5.04.1 A
Q5.04.2 B	Q5.04.3 B	Q5.03.1 D
Q5.03.2 B	Q5.03.3 A	Q5.02.1 D
Q5.00.1 C	Q5.99.1 A	Q5.98.1 C
Q5.98.2 A	Q5.96.1 *	Q5.95.1 B
Q5.95.2 A	Q5.94.1 C	Q5.93.1 B
Q5.92.1 0.3 to 0.3		

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